

1. What is a module? Explain the need of modules.

Answer:

A **module** is a Python file that contains functions, classes, variables, and other Python code. A module helps us organize and reuse code in different programs.

Example:

math is a built-in Python module that provides mathematical functions.

Need of modules:

1. **Code Reusability** – We can reuse the same code in different programs.
2. **Code Organization** – It helps to divide a large program into smaller parts.
3. **Easy Maintenance** – Changes can be made easily in a separate module.
4. **Avoids Code Repetition** – We do not need to write the same code again.
5. **Provides Ready-made Functions** – Python provides many built-in modules such as math, datetime, and random.

2. State the different ways of importing a module.

Answer:

There are different ways to import a module in Python:

1. Import the complete module

```
import math  
print(math.sqrt(25))
```

2. Import a specific function from a module

```
from math import sqrt  
print(sqrt(25))
```

3. Import multiple functions

```
from math import sqrt, factorial
```

```
print(sqrt(25))
```

```
print(factorial(5))
```

4. Import a module with an alias

```
import math as m
```

```
print(m.sqrt(25))
```

5. Import all functions and variables

```
from math import *
```

```
print(sqrt(25))
```